1. Paper Summaries

The following summaries draw on three peer-reviewed papers published between 2024 and 2025. Each covers the clinical or engineering problem, the method used, the reported outcome, and the authors' own stated limitation.

Tyler et al. (2024)

Emergency departments in the United States process approximately 131 million visits annually, yet conventional triage relies on the subjective judgement of the triage nurse, which varies across institutions, shift patterns, and individual experience. Tyler et al. conducted a scoping review of 29 peer-reviewed studies published between 2018 and 2023, identified through a systemised search of EMBASE, Ovid MEDLINE, and Web of Science, to examine whether machine learning could improve triage accuracy and consistency. Across all 29 studies, machine learning models outperformed conventional triage systems in predictive accuracy, disease identification, and risk assessment; one AI tool, KATE™, correctly assigned triage acuity in 75.7% of cases compared to 59.8% for triage nurses. The authors acknowledge that every study included in the review was retrospective and drawn from a single institution, which substantially limits how far the findings can be applied to other hospital settings or real-time clinical environments.

Da'Costa et al. (2025)

Overcrowding, resource constraints, and inconsistency in patient prioritisation place emergency departments under persistent pressure, and traditional triage systems — which depend heavily on subjective clinical judgement — deteriorate further during peak hours or mass casualty events. Da'Costa et al. conducted a narrative review of peer-reviewed articles published between 2015 and 2024, retrieved from PubMed, Scopus, IEEE Xplore, and Google Scholar, to examine the key components, benefits, and barriers of AI-driven triage systems. The review found that AI improves patient prioritisation and reduces wait times; one emergency department reported a 30% reduction in average wait time after integrating a real-time AI system that adjusted prioritisation automatically based on patient vitals and department capacity. However, the authors identify data quality issues, algorithmic bias, lack of clinician trust, and unresolved ethical concerns as significant barriers that continue to block widespread adoption.

Araouchi & Adda (2024)

An ageing population and growing case complexity place increasing strain on triage, which must prioritise patients by severity under conditions of limited resources and rising demand. Araouchi and Adda developed and evaluated TriageIntelli, a system built on seven machine learning models — including Support Vector Machines, Random Forests, Artificial Neural Networks, Gradient Boosting Machines, Logistic Regression, XGBoost, and a stacking ensemble — trained to predict patient triage levels using the Korean Triage and Acuity Scale. The stacking model, which combined the outputs of all seven algorithms, produced the strongest predictive performance, demonstrating that ensemble approaches consistently outperform any single model in triage classification. The authors note that the system was validated on a single dataset using one triage scale only, and that further testing across different health systems and clinical frameworks is required before the approach can be considered deployable at scale.

2. Gap Analysis

The three papers above converge on a clear picture: AI-assisted triage works in controlled retrospective settings, but consistent deployment in real clinical environments remains unsolved. Two specific gaps follow directly from the evidence.

Gap 1 — Absence of prospective, multi-site validation

Tyler et al. (2024) found that all 29 studies in their review were retrospective and conducted at single institutions. This is not simply a methodological footnote — it means no study has tested whether ML triage models hold their performance advantage when deployed prospectively, in real time, across different hospital environments. Araouchi and Adda (2024) make the same acknowledgement, noting that TriageIntelli was validated on a single dataset and cannot yet be considered generalisable. A 12-week pilot at Mercer General could address this gap directly by running an ML triage model in parallel with the existing system, logging real-time predictions against actual clinical outcomes, and producing the first prospective performance data from this institution.

Gap 2 — No validated mechanism for building clinician trust in AI triage

Da’Costa et al. (2025) identify lack of clinician trust as one of the primary barriers to AI triage adoption, yet none of the studies reviewed by Tyler et al. (2024) tested any structured approach to building or measuring that trust. The gap is not theoretical — it is operational. An AI system that clinicians distrust or override systematically cannot improve outcomes regardless of its accuracy in isolation. A 12-week pilot could address this by implementing a structured disagreement-logging protocol, in which clinicians record instances where they override the AI recommendation and provide a brief rationale. This produces both a trust baseline and actionable data for model refinement, which no existing study has attempted in a live ED setting.

3. Problem Statement

Emergency department triage remains inconsistent and resource-intensive, placing patients at risk through both over-triage and under-triage. The evidence is clear: machine learning models outperform human nurses in predictive accuracy. Yet no study has prospectively validated an ML triage model in a live clinical setting, and no structured mechanism exists to build the clinician trust that sustained adoption requires. Algorithmic bias and poor data quality compound the problem. This pilot will test a deployable, trustworthy AI triage system that bridges the gap between research performance and real-world clinical use.

4. References

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Tyler, S., Olis, M., Aust, N., Patel, L., Simon, L., Triantafyllidis, C., Patel, V., Lee, D. W., Ginsberg, B., Ahmad, H., & Jacobs, R. J. (2024). Use of artificial intelligence in triage in hospital emergency departments: A scoping review. Cureus, 16(5), e59906. https://doi.org/10.7759/cureus.59906